

1. Administrative & Study Identification

Full Study Title: Study Evaluating the Efficacy and Safety of Multiple Immunotherapy-Based Treatments and Combinations in Patients With Urothelial Carcinoma (MORPHEUS-UC) **Short Title/ Acronym:** MORPHEUS-mUC **Protocol Number:** W039613 **NCT Number:** NCT03869190 **Sponsor Name:** Roche (Hoffmann-La Roche) **Company:** Roche **Investigational Product:** Atezolizumab (monotherapy) and Atezolizumab combinations (with Magrolimab, Niraparib, Tocilizumab, Enfortumab vedotin, Linagliptin, or Isatuximab) **Phase of Development:** PHASE1, PHASE2 (Phase Ib/II) **Trial Status:** ACTIVE_NOT_RECRUITING **Medical Monitor:** Hoffmann-La Roche Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the early efficacy signals of multiple immunotherapy-based treatment combinations measured by investigator-assessed Objective Response Rate (ORR). **Secondary Objectives:** To evaluate investigator-assessed Progression-Free Survival (PFS), Overall Survival (OS), Duration of Response (DOR), Disease Control Rate (DCR), and the safety/tolerability of the novel combinations.

2.2 Background & Rationale Although cancer immunotherapy (CIT) has proven efficacy against multiple cancers, only subsets of patients have durable responses with monotherapy. CIT combination treatment targets multiple immune-evasion mechanisms and may thus be more likely to improve outcomes by simultaneously enhancing immune cell priming and activation, tumor infiltration, and tumor cell recognition. The MORPHEUS platform utilizes an umbrella design to simultaneously and efficiently identify early efficacy signals and evaluate the safety of multiple novel combinations in patients with locally advanced or metastatic urothelial carcinoma (mUC) progressing during or after a platinum-containing regimen.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Study) **Control Type:** Active Comparator (Atezolizumab Monotherapy) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Targeted Enrollment: 272 **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 01-Jun-2019 **Estimated Primary Completion:** Completed (approximate duration evaluated up to 5-7 years)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically documented, locally advanced or metastatic Urothelial Carcinoma (UC; including renal pelvis, ureters, urinary bladder, and urethra). 2. Disease progression during or following treatment with no more than one platinum-containing regimen for inoperable, locally advanced or metastatic UC, or disease recurrence. 3. Availability of a representative tumor specimen that is suitable for determination of PD-L1 and/or additional biomarker status by means of central testing.

4.2 Exclusion Criteria 1. Treatment with systemic immunosuppressive medication within 2 weeks prior to initiation of study treatment, or anticipation of need for systemic immunosuppressant medication during study treatment. 2. Active Epstein-Barr virus (EBV) infection or known or suspected chronic active EBV infection at screening (MIBC cohorts). 3. Impaired renal function or patients who decline neoadjuvant cisplatin-based chemotherapy (Cisplatin-Eligible MIBC Cohort).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multi-arm platform where participants are randomized to receive Atezolizumab control or an experimental arm (Atezolizumab + Magrolimab, Niraparib, Tocilizumab, Enfortumab vedotin, Linagliptin, or Isatuximab) on study-specific dosing schedules. **Route of Administration:** Intravenous (Atezolizumab and IV targeted agents) and Oral (for oral targeted combinations). **Duration of Treatment:** Baseline until disease progression or loss of clinical benefit.

5.2 Concomitant & Prohibited Therapy Permitted: Mineralocorticoids, corticosteroids for chronic obstructive pulmonary disease or asthma, or low-dose corticosteroids for orthostatic hypotension or adrenal insufficiency. **Prohibited:** Systemic immunosuppressive medications (unless a one-time pulse dose or acute low-dose is cleared by the Medical Monitor).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Specimen Collection (PD-L1/Biomarker central testing)
Baseline	Day 0	Randomization, First Dose
Interim	Regular Cycles	Safety Labs, Overlapping Toxicity Assessment, Efficacy Assessment (RECIST 1.1)
End of Study		Disease Progression Final Vital Signs, Exit Interview, IP Accountability, Transition to Stage 2 (if eligible)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR) for the mUC Cohort Stage 1; Pathological Complete Response (pCR) for Muscle Invasive Bladder Cancer (MIBC) Cohorts. **Measurement Method:** Investigator-assessed per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST 1.1).

7.2 Secondary & Exploratory Endpoints 1. Progression-Free Survival (PFS) 2. Overall Survival (OS) 3. Duration of Response (DOR) and Disease Control Rate (DCR)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection focusing on overlapping toxicities unique to the combinatorial immune checkpoint and targeted agent regimens. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Evaluates ongoing treatment arms to close existing arms that demonstrate minimal clinical activity or unacceptable toxicity, or open new treatment arms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy and Safety Set for the evaluation of adverse events. **Sample Size Justification:** Target sample of 272 is appropriately powered to detect early efficacy signals across multiple parallel Phase Ib/II combination arms against a shared control. **Handling of Missing Data:** Standard survival analysis methodologies (e.g., Kaplan-Meier) utilizing censoring for missing data points in OS and PFS.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring to assure data integrity across all global multicenter locations. **Audit Trail:** Data entered into Electronic Case Report Forms (eCRF) with central testing documentation for PD-L1/biomarker stratification.

11. Study Identifiers & Governance

Primary ID: W039613 **Registry Number:** NCT03869190 **Posting ID:** 943336558793472470 **Sponsor/Lead Organization:** Roche **Study Title:** MORPHEUS-UC **Official Regulatory Title:** A Phase Ib/II, Open-Label, Multicenter, Randomized Umbrella Study Evaluating the Efficacy

12. Clinical Framework (Urothelial Carcinoma Specific)

Disease Areas: Urothelial Carcinoma, Bladder Cancer **Primary Condition:** Urothelial Carcinoma (Locally Advanced or Metastatic) **Diagnosis Threshold:** Histologically documented progression during or following treatment with a platinum-containing regimen **Medical Methodology:** Cancer Immunotherapy (CIT) Combinations **Optimization Target:** Addressing immune-evasion mechanisms via PD-L1 inhibition combined with other targeted mechanism agents

13. Study Procedures & Methodology

Management Pathway: Umbrella trial stage progression (Stage 1 initial screening for efficacy; advancing to Stage 2 if signals meet the required threshold). **Education Protocol (HCP):** Site initiation visits and ongoing safety management for distinct CIT combination toxicities. **Education Protocol (Patient):** Informed consent detailing randomized combinatorial risks vs. monotherapy standard of care. **Quality Audit Mechanism:** Independent central review of tumor imaging and centralized biomarker assay evaluation.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 5-7 years **Onsite Visit Cadence:** Corresponds with combinatorial infusion/dosing cycles **Remote Monitoring Frequency:** Routine intervals tailored to cycle lengths **Checklist Review Interval:** Standard imaging RECIST 1.1 evaluations per protocol-specified timelines

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				